

CODE SPORT



In this month's column, we discuss some design concepts in natural language processing.

Design interview questions are getting popular in computer science interviews as a way of understanding the different concepts in machine learning/natural language processing (NLP)/system design. In last month's column, we discussed the design of a spell-checker system. In this month's column, we will discuss how to design a conversational task assistant that can support multiple common tasks for everyday use, such as calendar, reminders, ticket booking, restaurant booking, sports results querying, Web search etc. As in the case of any system design interview questions, the problem statement is typically sparse. The candidate is expected to ask clarifying questions to the interviewer to get a complete picture of the problem statement.

In system design interviews, there are two important skills the interviewers are looking for. The first is your ability to see the big picture – who the solution is designed for, how it will fit with the rest of the product, how it can be built and deployed, etc. The second is your deep expertise in designing specific technical details of the solution to make it highly reliable and fault-tolerant. These are two extreme end points of a spectrum of details. The first skill requires you to talk in detail about the entire system, whereas the second requires you to deep dive into specific implementation level details. Here the key is to start off with a high level overview of the solution, pointing out how it fits in with the larger picture. Starting from the targeted users, the key is to spell out how you would be able to identify success/failure criteria for your solution, not just intrinsically but in an end-to-end fashion.

As an illustrative example, for the above-mentioned problem of designing a conversational task assistant, you might be tempted to focus on NLP

models for intent classification or entity identification, and on how you can come up with models with better precision, recall and F1 score. Instead, first consider how the solution will fit in an overall product. While the F1 score could be a good intrinsic evaluation, what will be the external evaluation criteria for your solution? You need to talk about: (a) returning users to the conversational assistant; (b) measuring conversational quality; and (c) task fulfillment, as possible measures of extrinsic evaluation.

Identify who are the targeted users of the system. Will they be using one specific language or will multilingual support be needed? One important issue that needs to be highlighted is the fact that NLP tools and support for low resource languages are quite limited. For instance, if we assume that the solution needs to be deployed in a country like India where there are a plethora of languages, the NLP toolchain support for most of the Indic languages is pretty slim. There have been recent efforts around building NLP toolchains for Indic languages such as iNLTK and indicNLP. However, these are early efforts compared to the mature toolchains available for high resource languages such as Stanford coreNLP, spacy, NLTK, etc. The solution you design needs to be cognizant of this fact. Even when you can support different Indic languages, there are other challenges.

In a multilingual community, most people communicate in a bilingual fashion leading to most of the conversations being in code-switched or code-mixed text. Most of the standard NLP tasks such as POS tagging, named entity recognition, dependency parse, etc, are adversely impacted due to the presence of code switched text. One possibility is to preprocess the text through a multilingual machine translation system to convert the code switched text into

